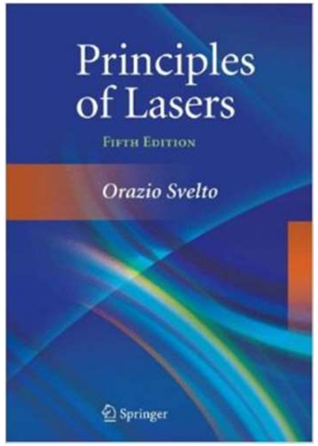
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch6-1
Control & Improvement of Laser Properties

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Chapter 6: Control & Improvement of Laser Performance

- ❖ 6.1 Mode Selection
  - ➢ Transverse mode selection
  - ➢ Longitudinal mode selection

- ❖ 6.2 Frequency-Stabilized Laser
  - ➢ External factors which influence the frequency stability
  - ➢ Frequency stabilization principle: optical length of stable cavity
  - ➢ Frequency stabilization method

6.1 Mode Selection

Purpose and Motivation: Fundamental transverse mode and single longitudinal mode possess very good spatial and temporal coherence.

❖ Transverse mode selection

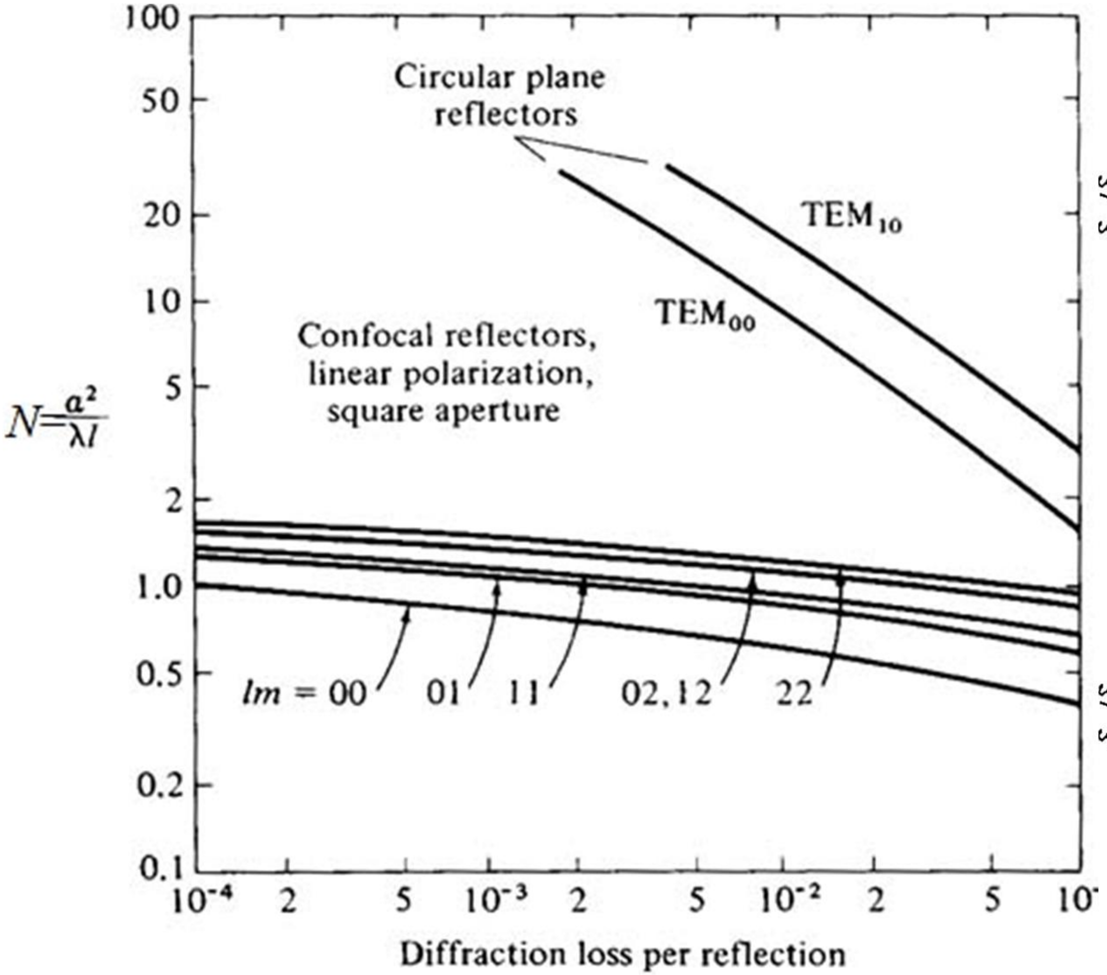
➢ Physics Foundation: different transverse modes have different diffraction losses

➢ Selection Rules

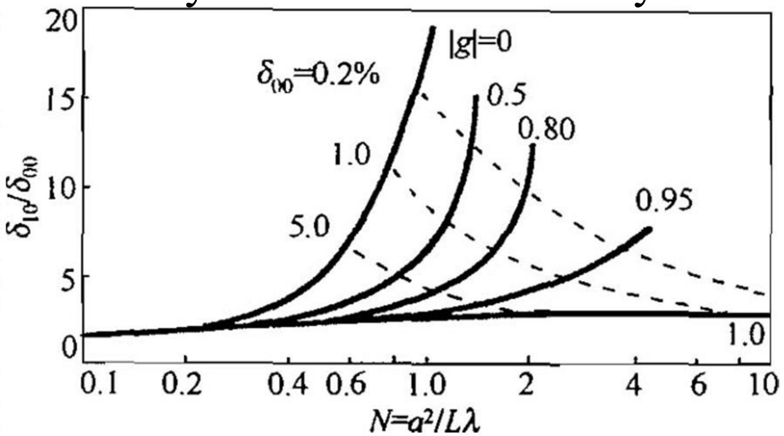
✓ Maximize the difference in diffraction loss between the fundamental and higher-order modes. Ensure that only the TEM_{00} mode meets the threshold condition, while suppressing all other modes.

✓ Minimize all losses except for diffraction loss, and maximize the proportion of diffraction loss in the total loss.

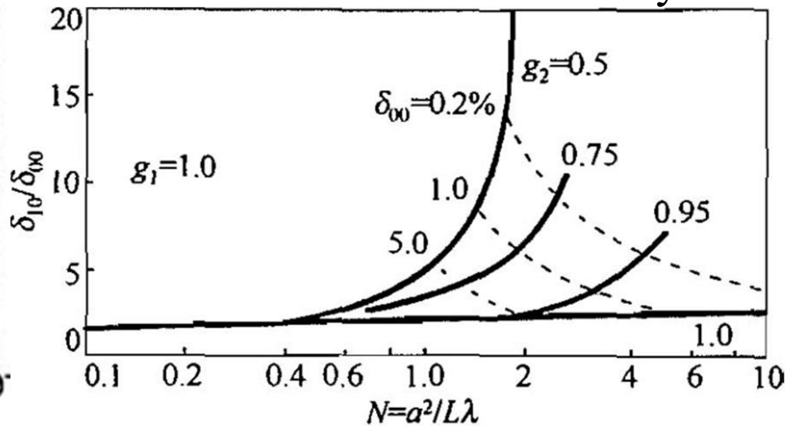
$$\alpha = \alpha_T + \alpha_d + \alpha_{s_1} + \alpha_{s_2} + \alpha_i$$



Symmetrical stable cavity



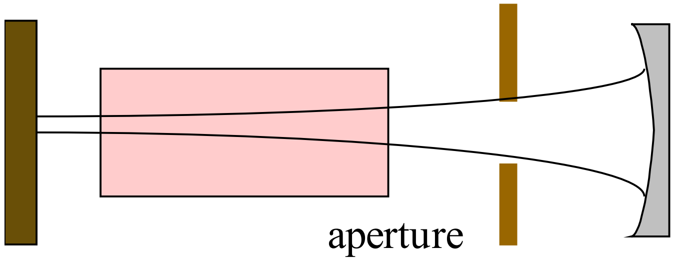
Flat concave stable cavity

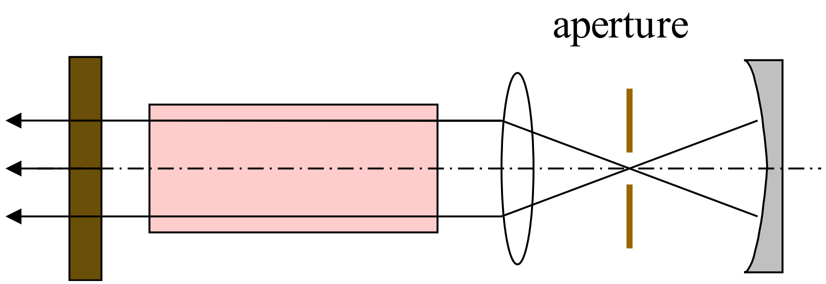


- Selection methods for transverse modes

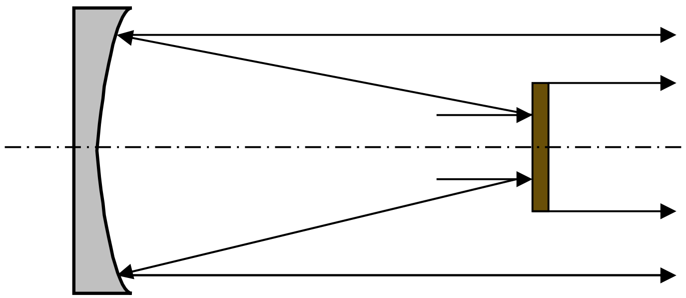
Cavity design; Small aperture; Nonstable cavity; Fine-tuned cavity

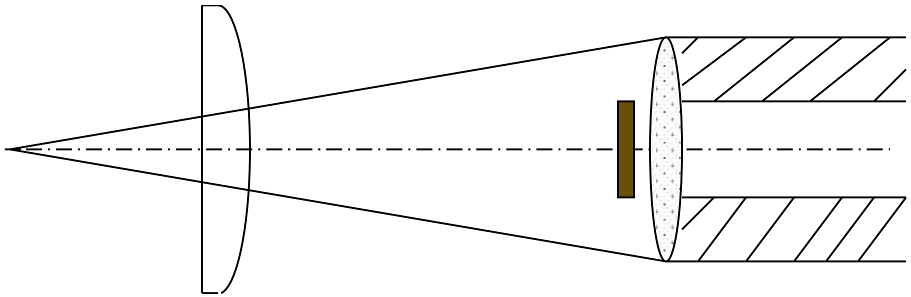
Small aperture for transverse mode selection





Nonstable cavity (Suitable for lasers with high gain)





Fine-tuned cavity

Cavity parameter g, N selection

❖ Longitudinal mode selection

- ✓ To achieve single longitudinal mode within the range of specific transition lines

➢ Selection rule

- ✓ Enlarge the gain difference between adjacent longitudinal modes or artificially induce loss difference

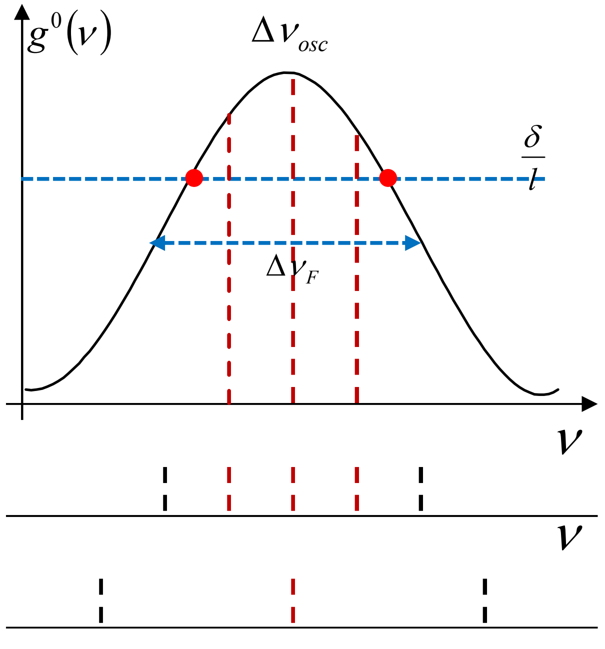
➢ Selection methods for longitudinal modes

- ✓ Short cavity

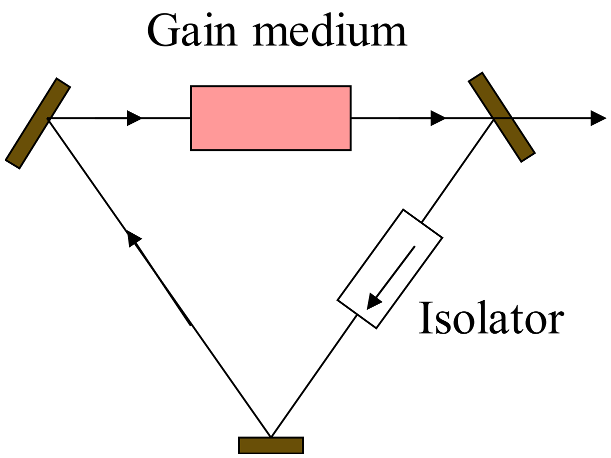
Decrease cavity length, increase the longitudinal mode spacing

$$\Delta\nu_q = \frac{c}{2L'} > \Delta\nu_{osc}$$

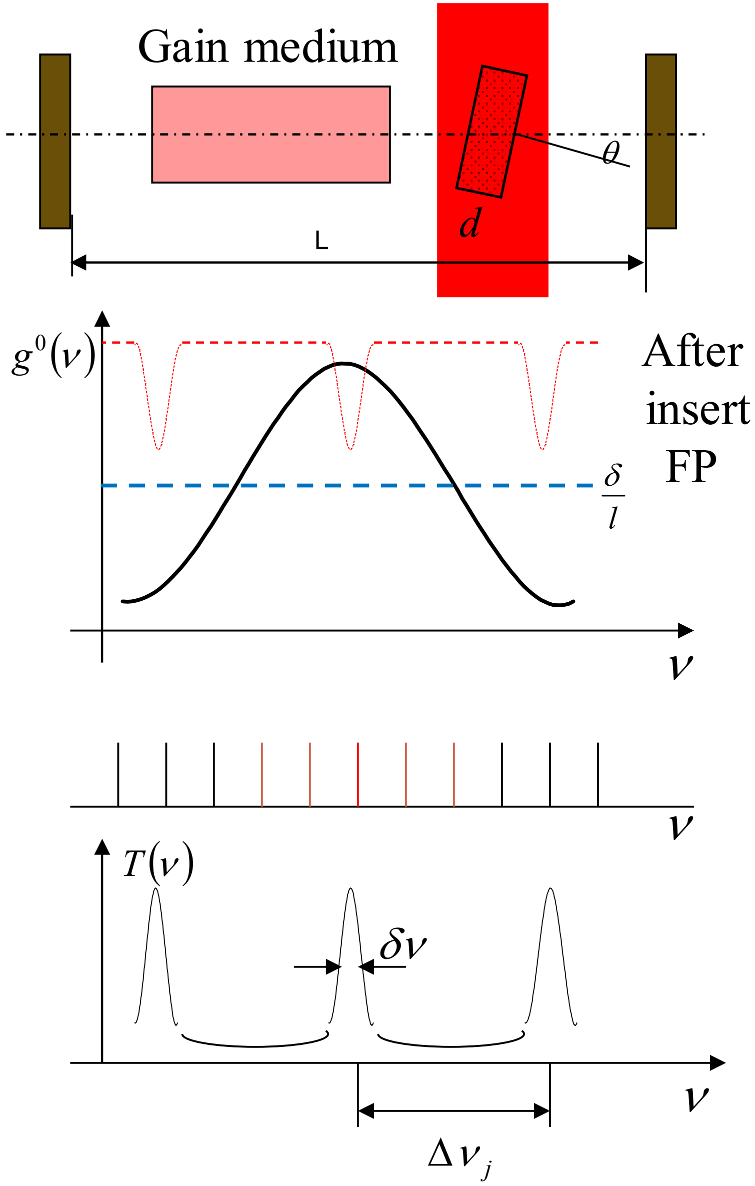
Suitable for lasers with narrow fluorescent linewidth, such as He-Ne laser



Travelling-wave cavity



F-P etalon



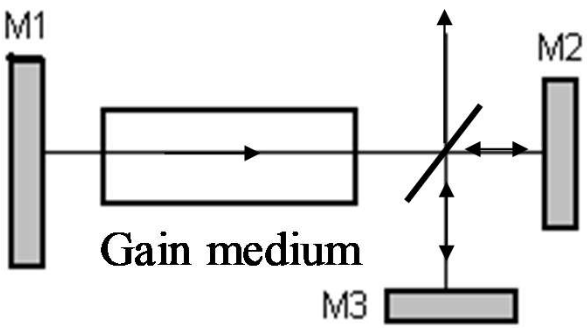
$$\Delta\nu_j = \frac{c}{2\mu d\cos\theta} \qquad \delta\nu = \frac{c}{2\pi\mu d}\frac{1-r}{\sqrt{r}}$$

Design requirement
for F-P etalon

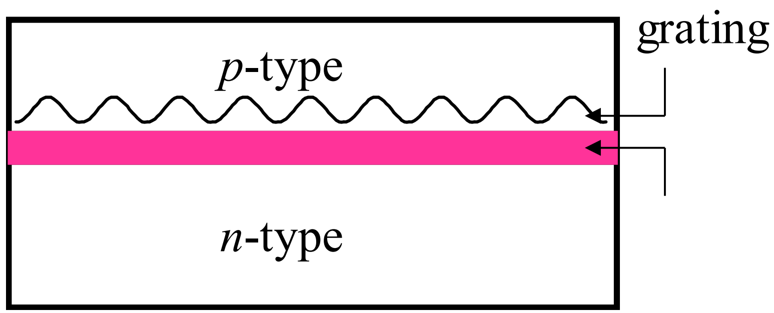
$$\Delta\nu_j > \Delta\nu_{osc}$$

$$\delta\nu < \Delta\nu_q = \frac{c}{2L'}$$

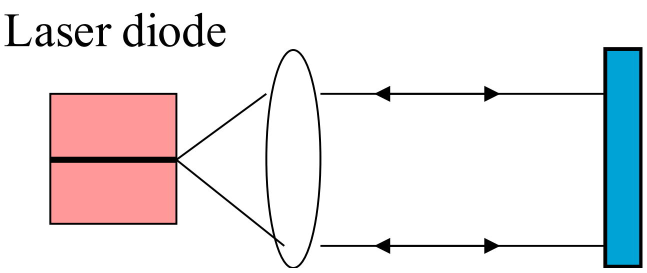
Combined cavity



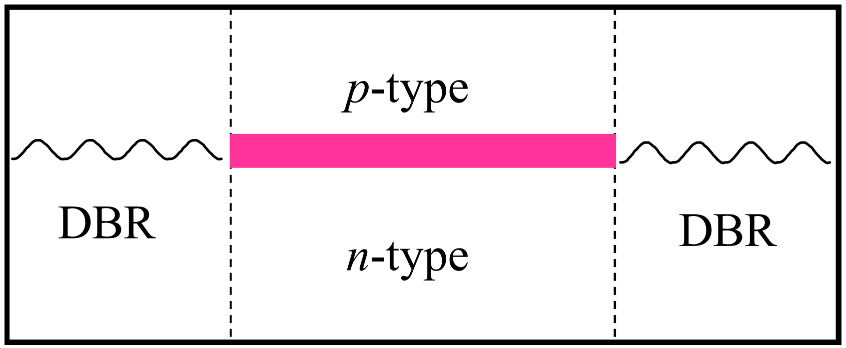
Distributed feedback (DFB) cavity



External cavity laser



Distributed Bragg Reflector (DBR)



6.2 Frequency-Stabilized Laser

❖ External factors which influence the frequency stability

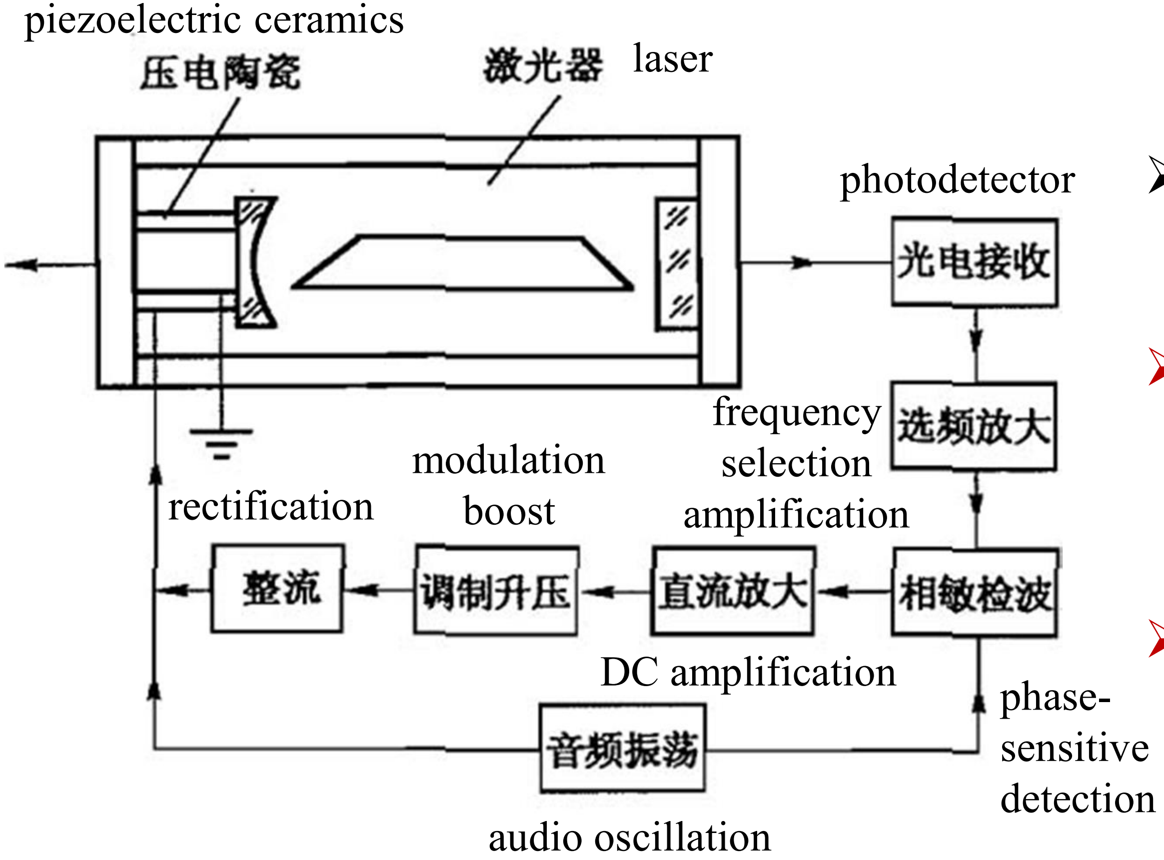
- Longitudinal frequency: $\boxed{\nu_q = q\frac{c}{2\eta L}}$ Frequency stability: $\frac{\Delta\nu}{\bar{\nu}}$
- Variation of cavity geometry: temperature and vibration ($10^{-3}/^{o}C$)
- Refractive index change: temperature, $\Delta\eta$ fluctuation (discharging and driving current, etc.), pressure, humidity
- Frequency stability of single mode He-Ne laser: 10^{-4}-10^{-5} ($\Delta\nu = 10^{10}$ Hz)
- Improved methods include constant temperature, shock proof, sound insulation, voltage stabilization, steady current : 10^{-7}

❖ Frequency stabilization principle: optical length of stable cavity

- Select standard reference frequency
- Error signal acquisition
- Drive electronic servo system to automatically adjust the cavity length

❖ Frequency stabilization method

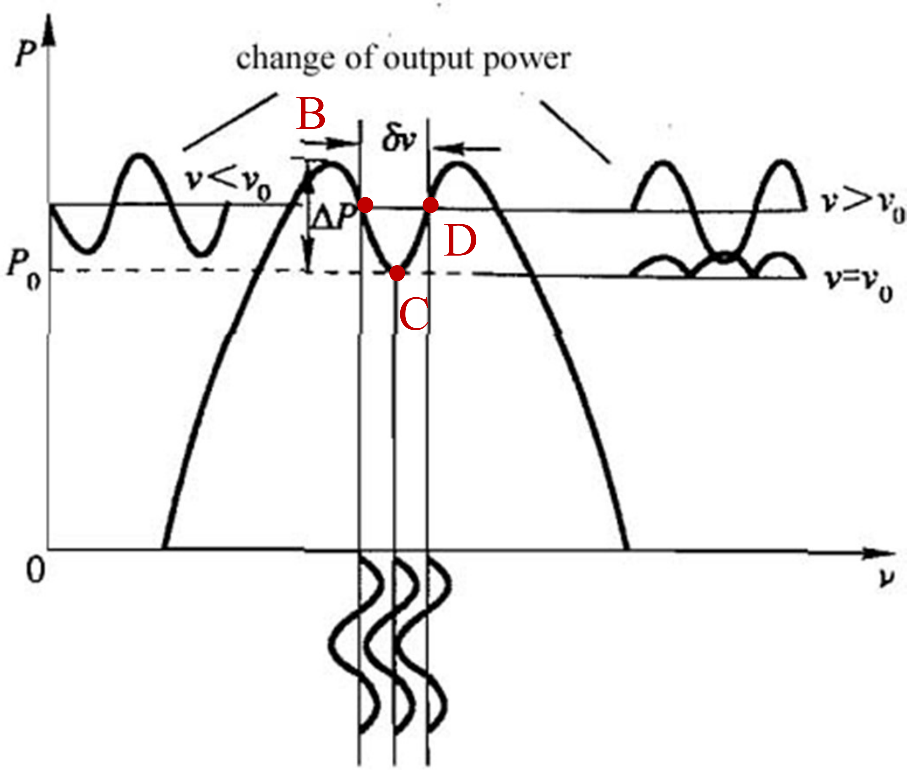
Lamp dip; Saturated absorption; Zeeman absorption; F-P etalon

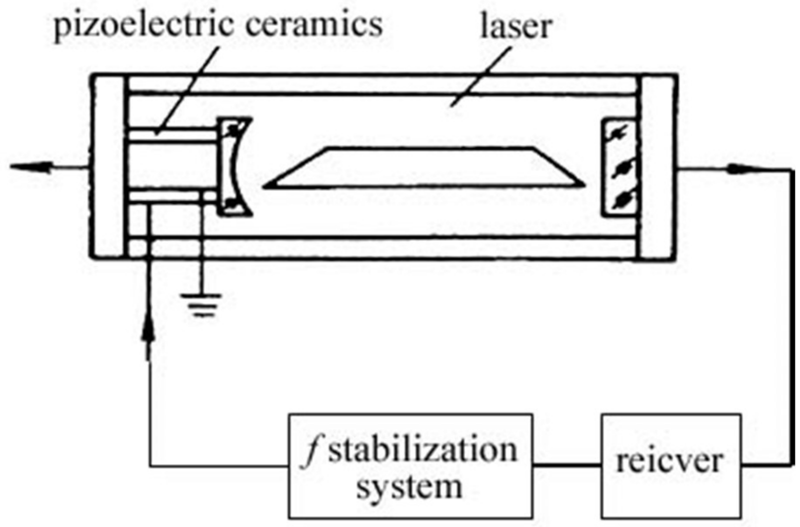


- Piezoelectric ceramics: change cavity length
- Direct-current: change the single mode laser output frequency
- AC signal: signal searching, judge positive or negative

forward voltage, external (+), internal (-), piezoelectric ceramics ↑→ cavity length↓→ ν_q↑

backward voltage, external (-), internal (+), piezoelectric ceramics ↓→ cavity length↑→ ν_q↓



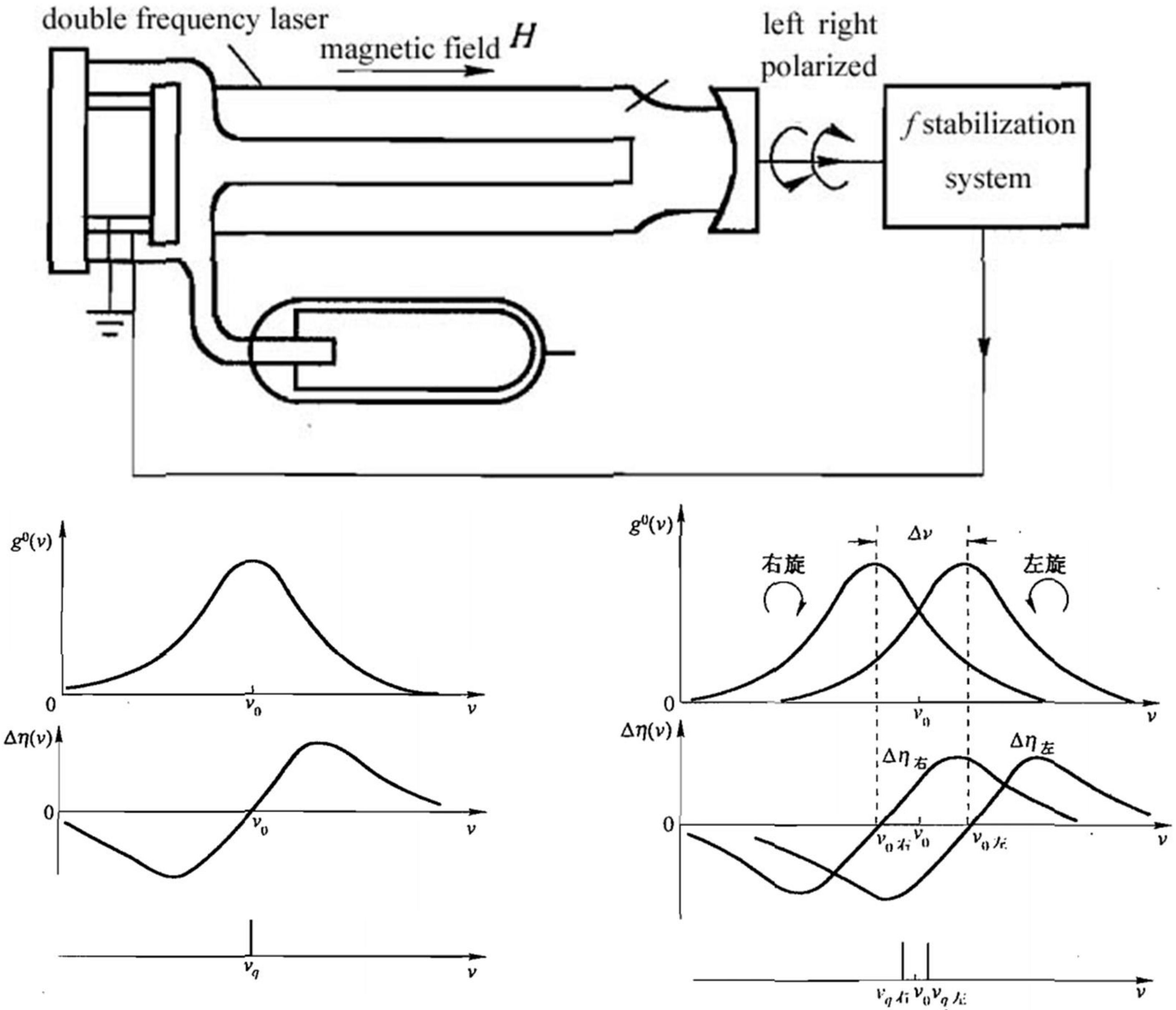


Frequency stabilization: 10^{-9}

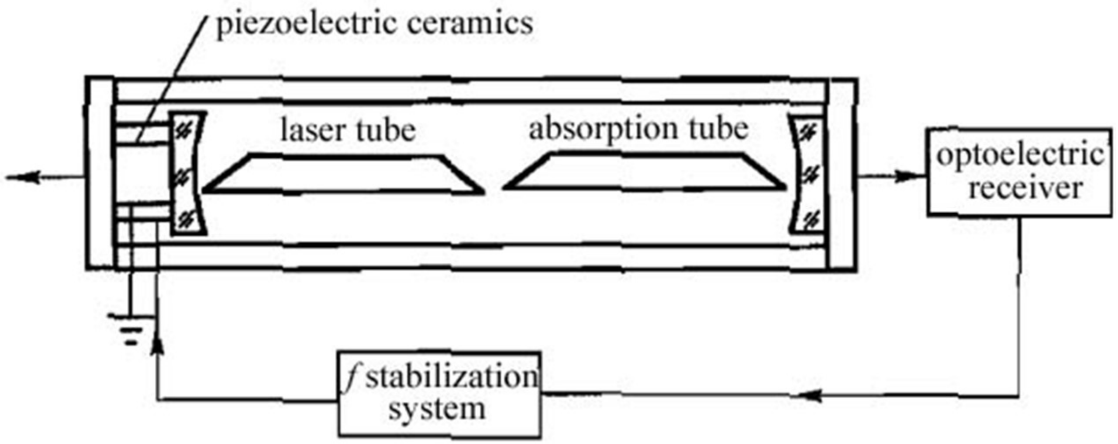
- $\nu > \nu_0$ D-point: same phase, reverse V, piezoelectric ceramics ↓→ cavity length↑→ ν_q ↓, pull back ν_0
- $\nu < \nu_0$ B-point: reverse phase, positive V, piezoelectric ceramics ↑ → cavity length↓→ ν_q ↑, pull back ν_0
- $\nu = \nu_0$ C-point: *2f*, 0 voltage, no change of piezoelectric ceramics, $\nu = \nu_0$

- ✓ Requires the symmetry of Lamb dip, narrow and dip (low pressure)
- ✓ Bad frequency reproducibility10^{-7}

- Zeeman frequency stabilization



- Saturated absorption-induced frequency stabilization: inverted Lamb dip



- Absorption tube pressure: 1~10 Pa
- Doppler broadening dominates
- Good frequency stability of absorption peak for low pressure gas

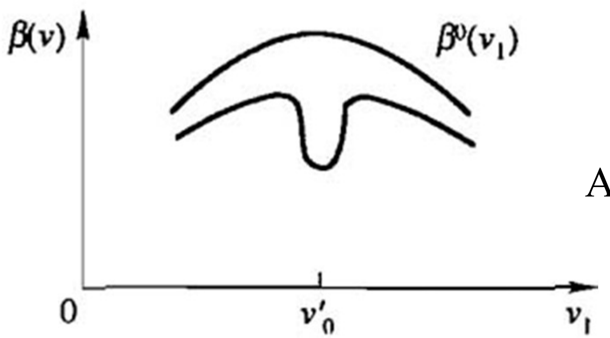
$$\nu_1 = \nu_0'$$

Absorption curve Hole-burning effect

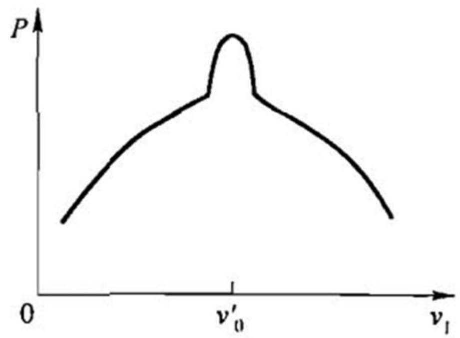
$$\beta(\nu_1) = \frac{\beta^0(\nu_0')}{\sqrt{1 + \frac{I_+ + I_-}{I_s}}}$$

Burning hole width

$$\delta\nu' = \sqrt{1 + \frac{2I_+}{I_s}}\Delta\nu_H'$$



Absorption saturation > gain saturation



Single pass loss of resonant cavity with absorption tube

$$\delta'(\nu_1) = \delta + \beta(\nu_1) L'$$

Output power $P = \frac{1}{2} A T I_s \left[\left(\frac{g_m l}{\delta'} \right)^2 - 1 \right]$

⇨ P vs. ν_1 curve forms an inverted lamb dip

- Frequency stabilization: 10^{-11}~10^{-12}
- Frequency reproducibility: 10^{-11}

  - ✓ 632.8 nm: isotopic iodine（碘）vapour
  - ✓ 3.39 μm: methane（甲烷）